

Changing Dimension in FFGFT

Geometric Reduction ($T^4 \rightarrow T^0$) and Representational Translation

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Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Abstract: $t \sim t + R_m$ "shown to be" an observable $\rightarrow$ treated as one; CHSH deviation $\sim \xi$ not resolvable with present hardware (Doc. 190 R142/R143).
- Time reading and conclusion: "currently just below resolution", "resolvable at $\sigma \sim 10^{-4}$ " $\rightarrow$ far below the hardware noise ($2\sqrt{2} - 2.74 = 0.088$, about 160Δ), not resolvable with present hardware.

Abstract

The reduction of the fundamental 4-torus structure T^4 of FFGFT to lower dimensions is examined as a formal chain $T^4 \rightarrow T^3 \rightarrow T^2 \rightarrow T^1 \rightarrow T^0$. The central claim is that this chain does *not* consist of steps of the same kind. It splits into two qualitatively distinct operations. The first step ($D4 \rightarrow D3$) is not a lossy projection but a *duality decompactification*: the compact mass dimension S_m^1 is unrolled into the non-compact time direction via the fundamental relation $\tilde{T} \cdot m = 1$. This operation preserves the mode content; only the direct visibility of compactness survives, as a discrete spectral signature encoded by the recursion ξ^n . The further steps ($D3 \rightarrow D2 \rightarrow D1 \rightarrow D0$) are genuine geometric projections of spatial directions: strictly lossy, not reversible, and *not* compensated by the recursion. The step $D3 \rightarrow D2$ is identified as a holographic projection. The topological identification $t \sim t + R_m$ is treated as a physical observable (T_{rev} , CHSH deviation $\sim \xi$, the latter not resolvable with present hardware). (*Corrected on 1 Oct. 2026: previously "shown to be a physical observable" without qualification – by Doc. 190 R142/R143 the ξ effect in CHSH is not resolvable with present hardware; the only documented hardware measurement (Doc. 147) confirms only the Bell violation.*) The downward reduction is contrasted with the *upward* operation: the bijective representational translation $T^4 \leftrightarrow$ Hilbert space/matrix (Type III, Doc. 230/206/231), which is *lossless* – precisely because it adds no geometric extra dimensions but expresses the same 4D content in a higher-dimensional language. The work thereby resolves the apparent tension between "projection loses information" and "compactified / decompactified", and classifies geometric downward reduction against representational upward translation.

1 Motivation: the tension in the question

FFGFT describes fundamental physics on a compact 4-torus T^4 with the single dimensionless parameter $\xi = 4/30000$ and the central relation $\tilde{T} \cdot m = 1$. The observable world, by contrast, is a three-dimensional space with time evolution, $\mathbb{R}^3 \times \mathbb{R}_t$.

Three observations on this transition appear to conflict:

- Loss picture.** Every projection $D \rightarrow D-1$ is not injective and therefore loses information. The chain $T^4 \rightarrow \dots \rightarrow T^0$ is not reversible.
- Compactness picture.** T^4 is fully compactified; the observable spacetime $\mathbb{R}^3 \times \mathbb{R}_t$ is decompactified.
- Upward picture.** There already exists a *bijective* translation of T^4 into a higher-dimensional space – the Hilbert space (Doc. 230) – which is *lossless*.

These three pictures cannot be naively superimposed: if the first step were pure loss (A), no *decompactified* time direction (B) could arise from it, since forgetting creates no

new dimension; and it would remain puzzling how an upward move (C) can be lossless while the downward move (A) destroys information.

The resolution: there is not one but *three* dimension-changing operations of *two kinds*. *Geometric* operations (Type I: mass-time duality; Type II: spatial projection) reduce the observable world out of T^4 ; the *representational* operation (Type III) translates T^4 upward without loss. The geometric-downward lossy versus representational-upward bijective move – this asymmetry is the central result of the work.

.2 Structure of T^4

Decomposition and radius classes

Definition .2.1 (Fundamental torus structure). The fundamental level of FFGFT is the compact 4-torus

$$T^4 = S_x^1 \times S_y^1 \times S_z^1 \times S_m^1,$$

with three *spatial* circles $S_{x,y,z}^1$ and one *mass circle* S_m^1 . All four components are compact.

Remark .2.2 (The fourth direction as substitute time). The fourth torus direction S_m^1 is not a mass dimension *alongside* a separate time, but the *substitute-time direction* itself. Mass and time are two *readings* of one and the same dimension; the fundamental relation $\tilde{T} \cdot m = 1$ is the translation dictionary:

$$\begin{array}{ccc} \underbrace{S_m^1 \text{ (compact)}}_{\text{mass reading, discrete spectrum}} & \xleftrightarrow{\tilde{T} \cdot m = 1} & \underbrace{\mathbb{R}_t \text{ (open)}}_{\text{time reading, continuous flow}} \end{array} .$$

There is no time “in addition” to the four directions; time is the decompact reading of the fourth direction (cf. Doc. 185). The two-type structure depends only on the separation of *small* and *large* compactification radii and is independent of the labelling.

The four directions carry two separated radius classes:

$$R_x = R_y = R_z \sim R_H \quad (\text{cosmological horizon, “large”}), \quad (1)$$

$$R_m \sim \xi \cdot \ell_P \quad (\text{Planck scale, “small”}). \quad (2)$$

Remark .2.3 (Status of the two scales). R_H here is the *declared external* (imported) cosmological scale, *not* derived from ξ . Intrinsic is only the *form* – the large/small radius-class split, equivalently the dimensionless ratio R_H/ℓ_P – not the absolute *factor* (Doc. 190, P20). The two-type structure of this document depends only on that class split, not on the numerical value of R_H . Conversely $R_m \sim \xi \ell_P$ sets the *top* of the KK tower ($1/R_m \sim \xi^{-1} m_P$, trans-Planckian); the observed particle masses lie further down the ladder via the recursion ξ^n (Doc. 159, 258/259).

Kaluza–Klein reading

A compact circle of radius R carries a discrete mass tower $m_n \sim n/R$, $n \in \mathbb{Z}$. The two radius classes therefore generate qualitatively different physics:

Circle	Radius	Mode tower
$S_{x,y,z}^1$	$R \sim R_H$ (large)	gap $\sim 1/R_H \approx 0$, quasi-continuous $\Rightarrow$ space
S_m^1	$R \sim \xi \ell_P$ (small)	gap huge, discrete and heavy $\Rightarrow$ mass spectrum

Proposition .2.4 ($\tilde{T}m = 1$ as compactification relation). *In natural units a compactification radius R has the dimension of a length or intrinsic time, $\tilde{T} \sim R$, while the generated mass scale is $m \sim 1/R$. The fundamental relation*

$$\tilde{T} \cdot m = 1$$

is thus structurally the Kaluza–Klein relation for the mass dimension.

.3 Two types of reduction

The *downward* direction (reduction of T^4) proceeds via two geometric operations, developed here. The lossless *upward* operation (Type III, representational translation) follows in Section 8.

Type I: duality decompactification (mass $\rightarrow$ time)

Definition .3.1 (Unrolling the mass dimension). The operation

$$S_m^1 \xrightarrow{\text{covering}} \mathbb{R}_t, \quad p : \mathbb{R}_t \rightarrow S_m^1, \quad t \mapsto e^{2\pi i t/R_m},$$

unrolls the compact mass circle into the non-compact time direction via its universal cover. It is governed by $\tilde{T} \cdot m = 1$.

Theorem .3.2 (Type I preserves mode content). *The unrolling $S_m^1 \rightarrow \mathbb{R}_t$ is an embedding, not a projection. Periodic structures on S_m^1 lift uniquely into $\mathbb{R}_t$; the function space on $\mathbb{R}_t$ is strictly larger. No mode content is forgotten; rather, an additional non-compact direction arises – time.*

Proof: (Sketch) The universal cover $p : \mathbb{R} \rightarrow S^1$ is a local homeomorphism and surjective. An R_m -periodic function on $\mathbb{R}$ corresponds bijectively to a function on S_m^1 (Fourier series, discrete spectrum). The reverse direction (compactification $\mathbb{R} \rightarrow S^1$) is lossy, since non-periodic parts do not descend. Unrolling therefore proceeds in the *embedding*, not the *projecting*, direction.

Corollary .3.3 (Time as embedding price). *The “price” of decompactification is not loss of information but the appearance of the time direction itself (Doc. 185). What does not remain directly visible is only the compactness: the periodicity of S_m^1 appears in the observable not as geometry but indirectly – as the discrete structure of the spectrum.*

Remark .3.4 (What “not totally loss-free” refers to). This is not about absolute loss but about *reading-dependent visibility*. In the mass reading the compactness is manifest – the spectrum is discrete and the identification $t \sim t + R_m$ is expressed directly. In the time reading the same compactness is only *latent*: from continuous time alone the topological identification is not uniquely reconstructible (the winding remains ambiguous). The duality guarantees full translatability of the spectrum; the identification is carried by the time reading only implicitly – through the discreteness of the spectrum. This is exactly where ξ^n acts.

What makes the unrolled time continuous and uniform

Two properties of the unrolled time have two different causes — and a third question (*that it unrolls at all*) has an entirely different answer.

Continuity: the cover. That $\mathbb{R}_t$ is *continuous* is a topological fact, not an addition: the unrolling $S_m^1 \rightarrow \mathbb{R}_t$ is the universal cover $p : \mathbb{R} \rightarrow S^1$ (Thm. .3.2), a local homeomorphism. The cover of a circle is the continuous line. Hence, as soon as a circle is unrolled, the result is necessarily continuous — and as an embedding, not a projection (no mode content is lost).

Uniformity: homogeneity and the fixed eigenvalue. That time flows at a *constant rate* rests on the homogeneity of the mass circle: S_m^1 is a $U(1)$, translation-invariant, no point distinguished. The constant, translation-invariant metric lifts to the cover $\mathbb{R}$ as an affine, uniform parametrization. The rate itself is the *single fixed eigenvalue* $m = 1/\tilde{T}$ (the rest energy), not a variable quantity. In the recursion picture: the constant multiplier ξ has a well-defined generator $\ln \xi$, so the discrete iteration ξ^n becomes the continuous dilatation flow $t = t_0 \xi^s$ (Doc. 285); constant multiplier $\Rightarrow$ constant rate.

That it unrolls at all: the measurement. Geometrically, in the compact T^4 all four circles are equal — there is no time. Not the geometry but the *measurement* (the embedded observer, Doc. 248) selects one circle as time and unrolls it; a physical act, not a mathematical limit (Doc. 285). In doing so the winding number is discarded.

The heart of it: continuity = discarded winding. The real answer to “what makes time continuous” is therefore the discarding of the winding. The winding is the discrete, compact information; in the time reading it is thrown away (the information-discarding step, Doc. 285), and the compactness becomes *latent* — from continuous time alone the winding is no longer uniquely reconstructible (Rem. .3.4). Continuous time is thus the circle with its winding forgotten; keep the winding and the discrete circle returns.

A subtlety: continuous exactly, uniform only up to $O(\xi)$. Two properties must be separated here. *Continuity* stays exact: $g_{\mu\nu} = \eta_{\mu\nu}(1 + \xi f)$ ($D_f = 3 - \xi$) is a *metric* correction, not a topological intervention — a bounded ξ -small modulation does not tear $\mathbb{R}$, no gaps arise. *Uniformity*, by contrast, holds only up to $O(\xi)$: the same ξ feedback that in the dynamic sector carries the AM sidebands (Doc. 286) makes the rate wobble by relative $\xi \approx 10^{-4}$; the flat metric η “ignores the fractal structure” (Doc. 051). The perfectly smooth, uniform time is thus the limit $\xi \rightarrow 0$ — the idealized time of relativity —, while the real FFGFT time is a continuum *with* a ξ -fine fractal rate modulation. Important: this is *not* discrete — fractal $\neq$ discrete; time remains an uncountable continuum, only not smoothly affine. (Whether this ξ graininess is the metric shadow of the latent compactness — a ξ remnant of the discarded winding — is a natural conjecture but not established: the latent compactness shows in the discrete spectrum (Rem. .3.4), the ξ correction in the metric.)

Type II: geometric projection (spatial)

Definition .3.5 (Forgetting a spatial direction). A projection $\pi : \mathbb{R}^k \rightarrow \mathbb{R}^{k-1}$ with $\ker(\pi) = \mathbb{R}$ forgets a spatial direction. It is not injective and therefore not invertible.

Theorem .3.6 (Type II is strictly lossy). *In a spatial projection $\mathbb{R}^k \rightarrow \mathbb{R}^{k-1}$ a full fibre $\mathbb{R}$ collapses. The information about the forgotten direction is not contained in the image and cannot be reconstructed without additional data (boundary conditions, holographic completeness). The recursion ξ^n does not act on this step.*

	Type I (mass→time)	Type II (spatial)
Operation	unrolling / covering	forgetting / projection
Direction	embedding	projecting
Mode content	preserved	lost
Reversibility	dual/spectral recoverable	not reversible
Role of ξ^n	spectral signature	ineffective
Residue	compactness only inferred	genuine geometric loss

.4 The formal chain, step by step

Step 1: $D4 \rightarrow D3 + \text{time}$ (Type I)

$$T^4 \longrightarrow \underbrace{T^3}_{\rightarrow \mathbb{R}^3} \times \underbrace{\mathbb{R}_t}_{\text{from } S_m^1}.$$

Step 1 bundles two information-preserving operations of *different* mechanism: (Ia) the *mass circle* unrolls to $\mathbb{R}_t$ via the duality $\tilde{T}m = 1$ per Theorem .3.2 (duality decompactification); (Ib) the three *spatial circles* decompactify through the *limiting process* of large radii ($R_{x,y,z} \rightarrow \infty$ for $r \ll R_H$) to $\mathbb{R}^3$ (limit decompactification). Both are embeddings, not projections; but (Ib) is *not* a duality step, it is a limit – see the clarification in Section .12, point 3.

Invariant: $\tilde{T} \cdot m = 1$ (Prop. .2.4); in the spatial limit, local flatness. *Loss:* none in mode content; only the visibility of compactness (Rem. .3.4).

Step 2: $D3 \rightarrow D2$ (Type II, holographic)

$$\pi_2 : \mathbb{R}^3 \longrightarrow \mathbb{R}^2, \quad \ker(\pi_2) = \mathbb{R} \text{ (depth direction).}$$

Invariant: area A . *Holography:* step 2 is mathematically the holographic principle; in the horizon case the Bekenstein–Hawking bound $S \leq A/(4\ell_P^2)$ (cf. Doc. 254/263). *Loss:* genuine depth direction.

Step 3: $D2 \rightarrow D1$ (Type II)

$$\pi_3 : \mathbb{R}^2 \longrightarrow \mathbb{R}, \quad \ker(\pi_3) = \mathbb{R}.$$

Invariant: length L . *Loss:* second spatial direction; no area measure remains.

Step 4: $D1 \rightarrow D0$ (Type II, total loss)

$$\pi_4 : \mathbb{R} \longrightarrow \{\text{pt}\}.$$

Invariant: only a discrete scalar measure ($n \in \mathbb{Z}$). *Loss:* the complete geometric structure.

.5 Reversibility, precise per type

Theorem .5.1 (Reversibility of the full chain). *Let $\Pi = \pi_4 \circ \pi_3 \circ \pi_2$ be the spatial subchain $\mathbb{R}^3 \rightarrow \{\text{pt}\}$ and D the Type I operation $S_m^1 \rightarrow \mathbb{R}_t$. Then:*

- (i) *D is, as a duality (Fourier/KK), invertible in mode content; the winding remains ambiguous. D is “almost reversible”.*
- (ii) *Π is not invertible: there is no continuous map $\rho : \{\text{pt}\} \rightarrow \mathbb{R}^3$ with $\rho \circ \Pi = \text{id}$.*

Proof: (Sketch) (i) follows from Theorem .3.2: unrolling is embedding, the discrete spectrum encodes the full periodic content; only the choice of lift (winding) is non-unique. (ii) follows from Theorem .3.6: each π_k has $\ker = \mathbb{R} \setminus \{0\}$. A left inverse would have to reconstruct the collapsed fibre structure; this information is not present in the image $\{\text{pt}\}$.

Corollary .5.2 (Resolution of the initial tension). *Pictures (A) and (B) are both correct but concern different operations. The non-reversibility (A) holds for the spatial projections (Type II). The decompactification to the time direction (B) is the embedding Type I step – it creates a dimension instead of forgetting one. There is no contradiction.*

.6 Role of the recursion ξ^n

The recursion $R : \xi^n \mapsto \xi^{n+1}$ is *not* a universal compensation for projection losses. Its function is narrowly bounded:

- **Type I:** ξ^n generates the discrete mass spectrum (particle ladder). This spectrum is the compact mode content of S_m^1 , dually expressed as time-frequency structure. ξ^n “fills in” nothing; it is the spectral signature through which the hidden compactness remains recognisable (Rem. .3.4).
- **Type II:** ξ^n has no effect. Spatial projection losses are genuine and untouched by the recursion.

Remark .6.1 (Correction of a tempting misrepresentation). A presentation that assigns the recursion a “compensation strength” decreasing monotonically across all four steps (full/partial/weak/none) is misleading. It presupposes a uniform lossy ladder that does not exist. The recursion belongs exclusively to the Type I step.

.7 Overview

Step	Operation	Type	Invariant	ξ^n
$D4 \rightarrow D3+t$	$S_m^1 \rightarrow \mathbb{R}_t$ (unroll)	I	$\tilde{T}m = 1$	spectral signature
$D3 \rightarrow D2$	forget space	II	area A	–
$D2 \rightarrow D1$	forget space	II	length L	–
$D1 \rightarrow D0$	forget space	II	$n \in \mathbb{Z}$	–

Table 1: The chain $T^4 \rightarrow T^0$ separated by type.

$$\underbrace{T^4 \longrightarrow \mathbb{R}^3 \times \mathbb{R}_t}_{\text{Type I: embedding, almost reversible}} \longrightarrow \underbrace{\mathbb{R}^2 \longrightarrow \mathbb{R} \longrightarrow \{\text{pt}\}}_{\text{Type II: projecting, not reversible}}$$

.8 The upward direction: representational translation (Type III)

The chain so far describes only the *downward* direction (reduction). There is a counter-direction – but it is *not* geometric.

Definition .8.1 (Type III: representational translation). The operation

$$T^4 \xrightleftharpoons{\text{bijective}} \mathcal{H}$$

translates the FFGFT mode formalism on T^4 into the (infinite-dimensional) Hilbert space $\mathcal{H}$, or the equivalent matrix algebra (Doc. 230, 206, 231). It is not an approximation but a *bijective structural statement*: every QM computation is expressible in FFGFT language and vice versa.

The dimensional asymmetry

Downward (geometric) and upward (representational) are different kinds of operation with *opposite* information properties:

- **Downward (Type II, geometric)**: lossy, not reversible – a fibre collapses (Theorem .3.6).
- **Upward (Type III, representational)**: bijective, lossless – it adds *no* physical dimensions but expresses the same 4D content in a higher-dimensional *language* in which it fits without reduction.

The losslessness follows *precisely from* the representational character – not from any gain of genuine extra dimensions.

Remark .8.2 (No geometrically higher dimensions). A geometric extension $T^4 \rightarrow T^5 \rightarrow \dots$ is excluded in FFGFT (Doc. 171): complexity arises from *fractal depth* ($D_f = 3 - \xi$, self-similarity), not from *more* dimensions. This is the deliberate contrast with string theory (10/11D, Calabi-Yau, landscape problem). “Higher” therefore always means Type III (representational) in FFGFT, never geometric.

FFGFT thus has three dimension-changing operations:

Type	Operation	Character	Information
I	mass $\leftrightarrow$ time ($\tilde{T}m = 1$)	geometric	almost reversible
II	spatial projection (down)	geometric	lossy
III	$T^4 \leftrightarrow \mathcal{H}$ /matrix (up)	representational	bijective

The bijectivity of Type III is the mirror image of the non-reversibility of Type II (Theorem .5.1): geometrically one loses on reducing, representationally one loses nothing on translating.

.9 Why exactly four circles? Minimal sufficiency, units, and visualization

The chain so far (Type I/II downward, Type III upward) presupposes the four-circle structure $T^4 = S_{x,y,z}^1 \times S_m^1$. This section answers *why* there are exactly four – deliberately *not* as a derivation of the number, but as *minimal sufficiency*.

3 + 1 is input, not derivation

No theory derives the dimension count of spacetime: the Standard Model and GR assume $3 + 1$; string theory fails on uniqueness (landscape problem). FFGFT, too, *does not derive the 4 forward*; it takes $3 + 1$ as *empirical input*. The question is therefore not “why is the world four-dimensional?” but “*given* $3 + 1$ – which structure is the minimal one that carries everything?”.

Minimal sufficiency: four instead of five

The duality saves one dimension

Naively one would need *five* building blocks: three for space, one for a mass spectrum (particles), one for time evolution. The fundamental relation $\tilde{T} \cdot m = 1$, however, identifies the *mass reading* and the *time reading* of *one* dimension (Type I, Theorem .3.2): the fourth circle S_m^1 carries the mass spectrum as its KK tower *and*, decompactified, time. The duality lets *one* circle do the work of two – so the naive 5 collapses to **4**.

- **Fewer** (only T^3): no mass spectrum, no time.
- **Four** (T^3 + one dual circle): space, masses *and* time from *one* object, without redundancy.
- **More** (a second such circle): a second independent mass tower or a second time – superfluous, without correlate.

“Optimal” therefore means *minimal-sufficient*: the smallest dimension count in which space, the full mass spectrum, and time jointly emerge from one geometric object.

The four circles as units bookkeeping

The four-circle structure *is* at the same time the exponent bookkeeping of the SI conversion factors – each circle type carries its bridge:

$$\underbrace{S_{x,y,z}^1}_{\text{space}} \xrightarrow{c} \text{time}, \quad \underbrace{S_m^1}_{\text{mass}} \xrightarrow{\hbar (\tilde{T}m=1)} \text{time}.$$

c is the space-time bridge of the three spatial circles, $\hbar$ the mass-time bridge of the mass circle. This is why the SI conversions are *indispensable* once one wants to determine the exponent structure correctly:

Remark .9.1 (Why natural units hide the powers). Natural units ($c = \hbar = 1$) collapse L , T and M to *one* axis. This removes precisely the distinction that carries the exponent structure – the independent *power of c* resp. *of time*. c and $\hbar$ may be set to 1 only *after* the exponents have been read off; before that the information is destroyed. The factors are the scaffolding that keeps the powers visible – and easy to handle precisely because their whole role is that scaffolding without a free parameter. G , too, is *derived* – reconstructed from ℓ_P , c , $\hbar$, for which it needs only the SI relations (Docs. 012/056) – and carries *no* free factor. The only free factor sits not in a conversion or natural constant but in the *cosmological absolute scale* R_H (exponent 41/4; Doc. 190, R20).

The same structure as Hilbert visualization

The four-circle geometry is also the *faithful* visualization of the infinite-dimensional Hilbert space (Type III, Def. .8.1) – two readings of *the same* bijection:

- **generative/ontological**: $T^4 \rightarrow \mathcal{H}$ – the geometry *generates* the algebra (Doc. 206), not the reverse;
- **visualizing/epistemic**: the abstract $\mathcal{H}$ becomes intuitable at all only through the 4D geometry.

“Higher-dimensional” (the ∞ -dimensional $\mathcal{H}$) does *not* mean “more fundamental”: the generative direction is $T^4 \rightarrow \mathcal{H}$. Precisely *because* the 4D structure is generative (not a mere picture), the visualization is *exact* – bijective, not a cartoon.

Summary of the dimension line

What "optimal" means

T^4 resp. $T^3 + 1$ time is the *minimal-sufficient, dual-economical and Hilbert-faithful* configuration – *given* $3 + 1$. The strength lies in the economy (one dimension saved by $\tilde{T}m = 1$) and the faithfulness (lossless Hilbert representation), *not* in a uniqueness of the number. The bare 4 is input, not a forward result (Doc. 190, R34) – the same signature as for the ξ -magnitude (R33) and the cosmological scale (R20): *structure forward, number input*.

.10 In the compact view, time is just another circle: the asymmetry is the measurement

In the compact representation $T^4 = S^1 \times S^1 \times S^1 \times S^1$ all four circles are identical: dimensionless cyclic coordinates, no signature, no space/time distinction. Time is written *exactly* as the three spatial circles; at the compact level the Lorentzian asymmetry does not exist. This is why, in FFGFT, the time coordinate is simple to write – it is one circle like the others. The complication enters only when one leaves the compact picture.

The asymmetry (one time, signature $(3, 1)$) appears only under decompactification (Type I, Def. .3.1): exactly one circle – the mass circle, $\tilde{T} \cdot m = 1$ – unrolls through its universal cover into non-compact time, while the other three stay compact (discrete spectrum) or project (space, Type II).

Which circle becomes time is therefore not an intrinsic-geometric fact to be derived. The geometry is symmetric, and nothing in it should single one direction out. Time is *relational*: the selection is the measurement itself. The decompactification *is* the measurement – the read-out/projection of this document and of Doc. 284 – and committing one circle to non-compact, experienced time is exactly the information-discarding step a measurement performs. The embedded observer, reading out from the inside (Doc. 248, self-referentiality), experiences one direction as duration; that experience constitutes the time circle. The symmetry-breaking is the measurement, not the geometry.

[The time circle is selected by the measurement] In the compact view the four circles of T^4 are symmetric. Time is not chosen by the geometry but is relational: the symmetry-breaking *is* the measurement (the decompactification of Type I). Three statements then close the bookkeeping with no gap:

- **That there is one time** (not two) is the signature/predictivity constraint – one timelike direction gives hyperbolic field equations with a well-posed Cauchy problem; two or more give ultrahyperbolic equations with no well-posed initial-value problem.
- **Which circle is time** is fixed by the measurement, i.e. by the experienced situation of the embedded observer (Doc. 248).
- **That the split is $3+1$** is the structure of being an embedded observer – the empirical input of Doc. 190 (P34), read here as the observer's situation rather than a free parameter.

The contrast with a non-compactified treatment of time is then sharp. A framework that writes time as a complicated structure built on already-physical time t (for instance a spiral-time operator $\psi(t) = t + i\varphi(t) + j\chi(t)$, Doc. 283/284) has *reified* the result of the measurement into its kinematics, instead of recognising experienced time as the measurement-induced

decompactification of a symmetric circle. In the compact view that complication is unnecessary: time is one more circle, and only the read-out turns it into the non-periodic axis we experience.

.11 Physical testability of the identification $t \sim t + R_m$

The topological identification $t \sim t + R_m$ is not a mere accompanying structure of the duality but a physical observable. In the time reading its period is *exactly* the fundamental revolution time

$$T_{\text{rev}} = \frac{R_m}{c} = \frac{\xi \ell_P}{c} \approx 7.2 \times 10^{-48} \text{ s}, \quad \Delta\phi = 2\pi\xi \approx 8.4 \times 10^{-4} \text{ rad/revolution},$$

so that $t \sim t + R_m$ is identical with $t \sim t + cT_{\text{rev}}$ (Doc. 183/185). The compactness of the fourth direction *is* the revival time.

Two testable readings

- **Mass reading (mature, quantitative).** The discrete mode spectrum generates the particle masses as torus overtones with step $1/D$; from this follow Koide, $g-2$ and the three generations (Doc. 159, 258/259). Here the compactness is numerically realised.
- **Time reading (falsifiable in principle, not resolvable with present hardware).** The periodicity produces a CHSH deviation from the Tsirelson bound of order ξ . For 73 qubits FFGFT predicts $\Delta \approx 5.4 \times 10^{-4}$. The hardware test IBM Heron r2 (May 2026) confirms the Bell violation ($S = 2.74$) and the executability of T0 circuits; the ξ effect, however, lies far below the hardware noise and is not resolvable with present hardware (Doc. 147). *(Corrected on 1 Oct. 2026: previously "currently below resolution", "resolvable at $\sigma \sim 10^{-4}$ " and "not yet resolvable" – $\sigma \sim 10^{-4}$ is only the statistical bound; the measured deviation from the Tsirelson bound is $2\sqrt{2} - 2.74 = 0.088$ (3.1%), about 160 times Δ ; by Doc. 190 R142/R143 the ξ effect is not resolvable with present hardware.)*

Scaling: linear versus logarithmic

Regime dependence of phase accumulation

A naive *coherent linear accumulation* $r \cdot \Delta\phi$ over r revolutions would yield physically absurd phase drifts (of order 10^{44} rad/s) and contradict working atomic clocks and deep quantum circuits. The lab-testable prediction therefore scales *logarithmically*,

$$\delta\phi \sim \frac{\xi \ln(r)}{D_f} \approx 2 \times 10^{-4} \quad (D_f = 3 - \xi),$$

not linearly. The linear form $r \cdot \Delta\phi$ of Doc. 185 is the *worst-case bound of the RSA argument* (coherence over $r \approx N$ cycles), not the lab prediction. This regime separation explains why deep coherent circuits can run while the RSA factorisation of large keys remains geometrically protected.

Conclusion. $t \sim t + R_m$ is carried in the FFGFT corpus as an observable (T_{rev} , CHSH deviation $\sim \xi$): quantitatively confirmed in the mass reading, falsifiable in principle in the time reading, but not resolvable with present hardware (far below the hardware noise). *(Corrected on 1 Oct. 2026: previously "currently just below experimental resolution" –*

see the note on the time reading above; Doc. 190 R142/R143.) It is not an unfalsifiable accompanying structure.

.12 Classification and open questions

Connections. Doc. 185 (Time as the Embedding Price) provides the ontological basis of the Type I step; Doc. 263 (holographic projection) and Doc. 254 (area-bound theorem) the framework for step 2; Doc. 206 (Triangle–Matrix reduction) confirms that the 4D level is generative; Doc. 147/183 provide the time-side testability. The upward operation (Type III) is developed in Doc. 230 (bijective Hilbert-space translation), Doc. 231 (necessary extensions) and Doc. 206 (matrix representation); Doc. 171 justifies the exclusion of geometrically higher dimensions.

Open points.

1. **Structural assumption – confirmed.** The decomposition $T^4 = S_{x,y,z}^1 \times S_m^1$ with S_m^1 as substitute-time direction and $\tilde{T} \cdot m = 1$ as translation is fixed (Rem. .2.2).
2. **Linear/logarithmic scaling.** The linear worst-case bound $r \cdot \Delta\phi$ (Doc. 185, RSA regime) and the logarithmic lab prediction $\xi \ln(r)/D_f$ (Doc. 147) should be marked explicitly as regime-dependent to secure consistency of presentation.
3. **Spatial limit – clarified.** The decompactification $S_{x,y,z}^1 \rightarrow \mathbb{R}^3$ is a *limiting process* ($R_{x,y,z} \rightarrow \infty$), neither the Type I duality nor a Type II projection, but a distinct *limit-type* information-preserving operation (Ib). It is therefore carried in Step 1 explicitly apart from the mass-circle duality (Ia); both share the embedding character, not the mechanism.
4. **Holographic completeness.** To what extent can the Type II loss at $D3 \rightarrow D2$ be reconstructed exactly (not just partially) from boundary data? Connection to Doc. 254/263.